

Social Network Analysis Using a Graph Database

Aarthee A

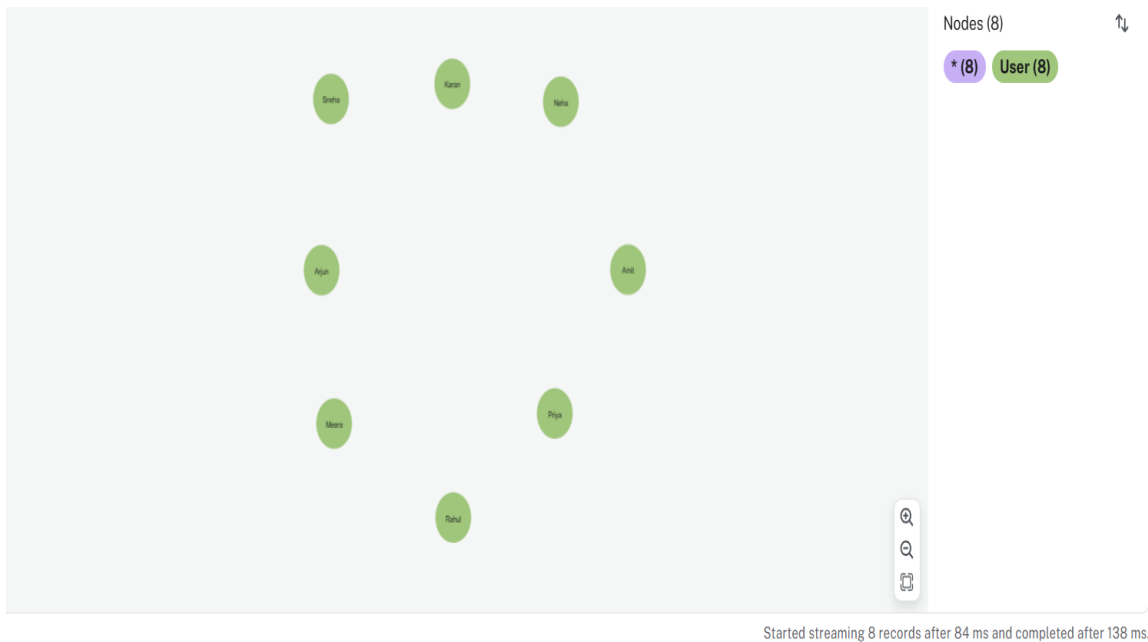
25030242001

1. User Attributes

User	Age	City	Profession	Activity	Influence
Rahul	29	Pune	Data Scientist	92	87
Priya	27	Mumbai	AI Engineer	88	82
Amit	31	Pune	Entrepreneur	76	79
Neha	25	Delhi	Researcher	95	91
Karan	30	Pune	Software Engineer	69	73
Sneha	28	Mumbai	Product Manager	84	86
Arjun	26	Bengaluru	ML Engineer	91	89
Meera	32	Delhi	Consultant	72	77

2. Create the Users

```
CREATE
(r:User {name:'Rahul', age:29, city:'Pune', profession:'Data Scientist',
activityLevel:92, influenceScore:87}),
(p:User {name:'Priya', age:27, city:'Mumbai', profession:'AI Engineer',
activityLevel:88, influenceScore:82}),
(a:User {name:'Amit', age:31, city:'Pune', profession:'Entrepreneur', activityLevel:76,
influenceScore:79}),
(n:User {name:'Neha', age:25, city:'Delhi', profession:'Researcher', activityLevel:95,
influenceScore:91}),
(k:User {name:'Karan', age:30, city:'Pune', profession:'Software Engineer',
activityLevel:69, influenceScore:73}),
(s:User {name:'Sneha', age:28, city:'Mumbai', profession:'Product Manager',
activityLevel:84, influenceScore:86}),
(ar:User {name:'Arjun', age:26, city:'Bengaluru', profession:'ML Engineer',
activityLevel:91, influenceScore:89}),
(m:User {name:'Meera', age:32, city:'Delhi', profession:'Consultant', activityLevel:72,
influenceScore:77});
```



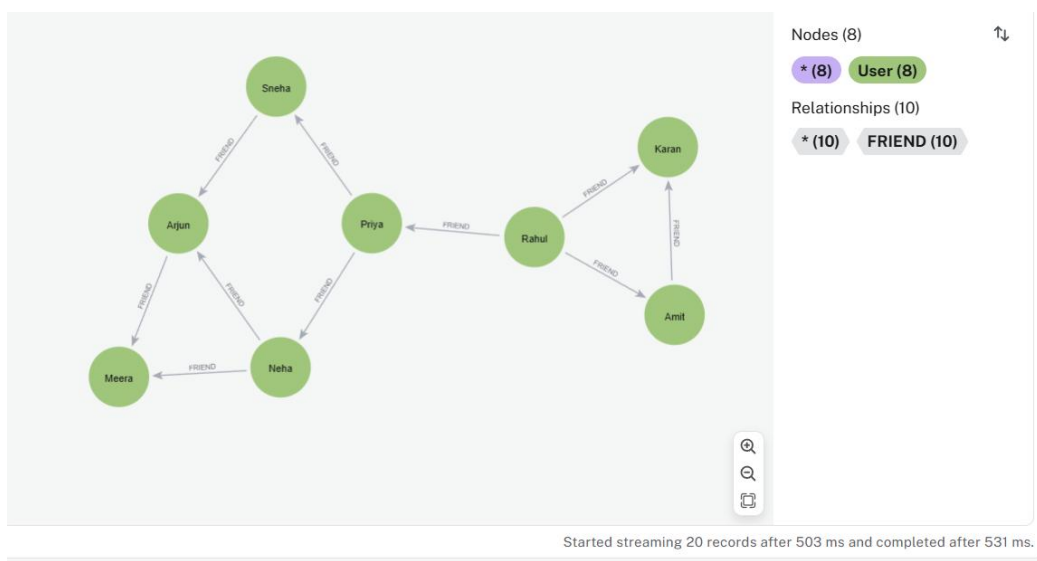
3. Create Friendship Relationships

MATCH

```
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (a:User {name:'Amit'}), (n:User {name:'Neha'}),
(k:User {name:'Karan'}), (s:User {name:'Sneha'}), (ar:User {name:'Arjun'}), (m:User {name:'Meera'})
```

CREATE

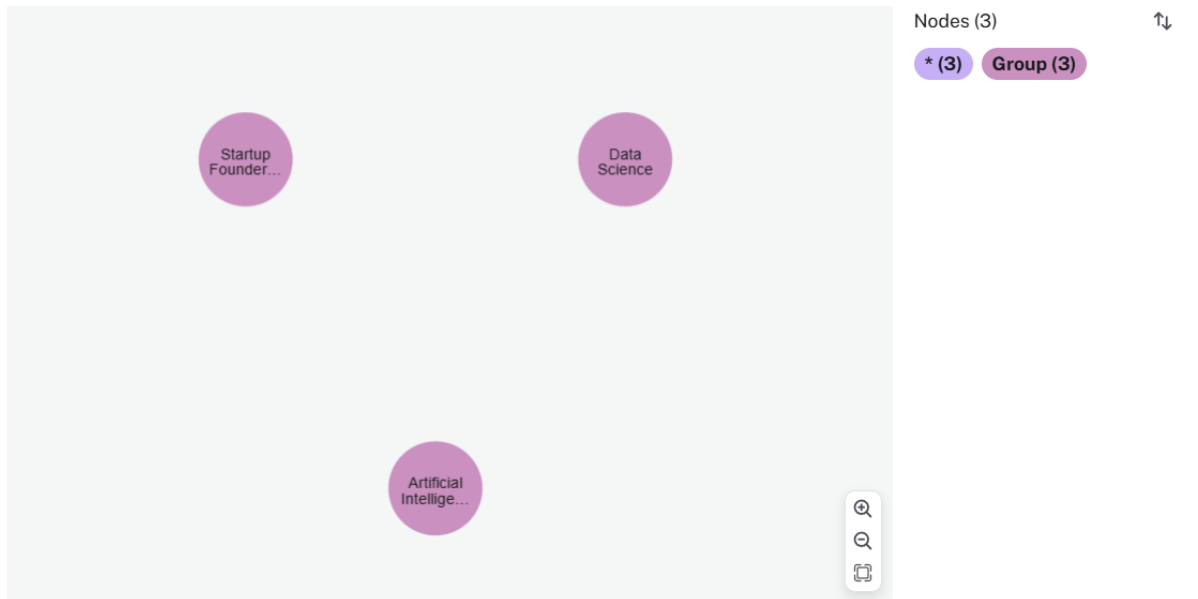
```
(r)-[:FRIEND {since:2021, strength:0.91}]->(p),
(r)-[:FRIEND {since:2022, strength:0.75}]->(a),
(r)-[:FRIEND {since:2023, strength:0.64}]->(k),
(p)-[:FRIEND {since:2022, strength:0.83}]->(n),
(p)-[:FRIEND {since:2021, strength:0.89}]->(s),
(n)-[:FRIEND {since:2023, strength:0.71}]->(ar),
```



```
(n)-[:FRIEND {since:2022, strength:0.68}]->(m),
(a)-[:FRIEND {since:2024, strength:0.62}]->(k),
(s)-[:FRIEND {since:2023, strength:0.77}]->(ar),
(ar)-[:FRIEND {since:2024, strength:0.66}]->(m);
```

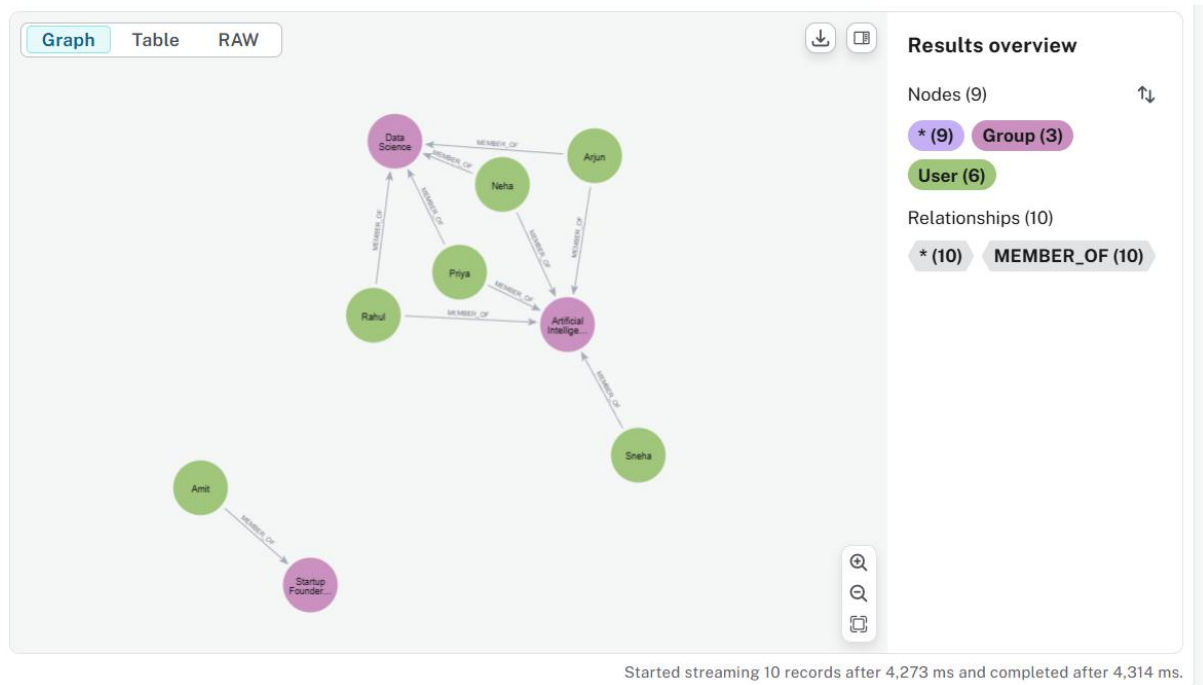
4. Create Groups

```
CREATE
(g1:Group {name:'Artificial Intelligence', category:'Technology', members:420}),
(g2:Group {name:'Data Science', category:'Technology', members:350}),
(g3:Group {name:'Startup Founders', category:'Business', members:210});
```



5. Connect Users to Groups

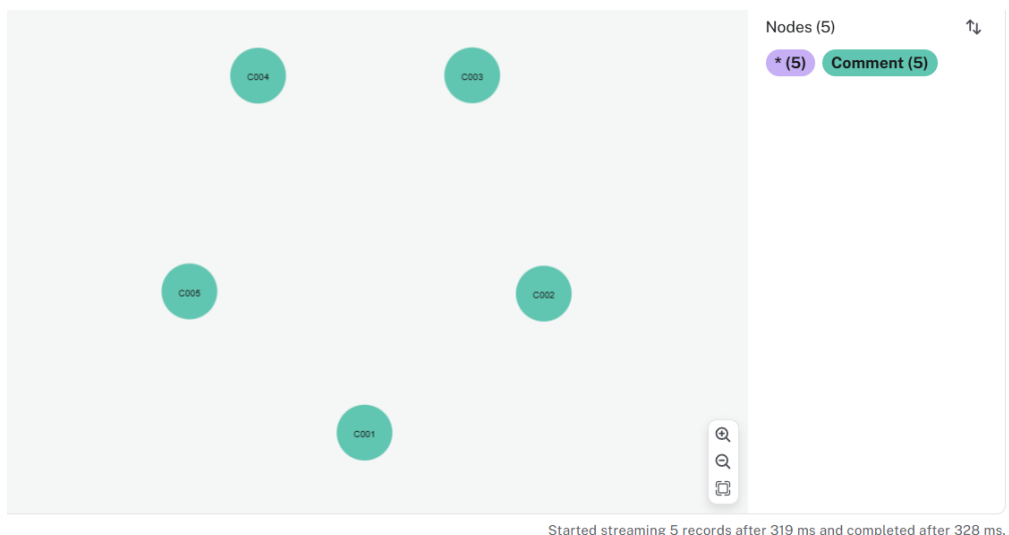
```
MATCH
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (n:User {name:'Neha'}), (a:User {name:'Amit'}),
(s:User {name:'Sneha'}), (ar:User {name:'Arjun'}),
(g1:Group {name:'Artificial Intelligence'}), (g2:Group {name:'Data Science'}),
(g3:Group {name:'Startup Founders'})
CREATE
(r)-[:MEMBER_OF {role:'Member', since:2022}]->(g1),
(r)-[:MEMBER_OF {role:'Moderator', since:2021}]->(g2),
(p)-[:MEMBER_OF {role:'Member', since:2023}]->(g1),
(p)-[:MEMBER_OF {role:'Member', since:2022}]->(g2),
(n)-[:MEMBER_OF {role:'Researcher', since:2021}]->(g1),
(n)-[:MEMBER_OF {role:'Member', since:2023}]->(g2),
(a)-[:MEMBER_OF {role:'Founder', since:2022}]->(g3),
(s)-[:MEMBER_OF {role:'Member', since:2023}]->(g1),
(ar)-[:MEMBER_OF {role:'Member', since:2022}]->(g1),
(ar)-[:MEMBER_OF {role:'Member', since:2023}]->(g2);
```



6. Create Comments

Comments are modeled as independent nodes so each comment can have its own sentiment, toxicity, likes, timestamp and topic.

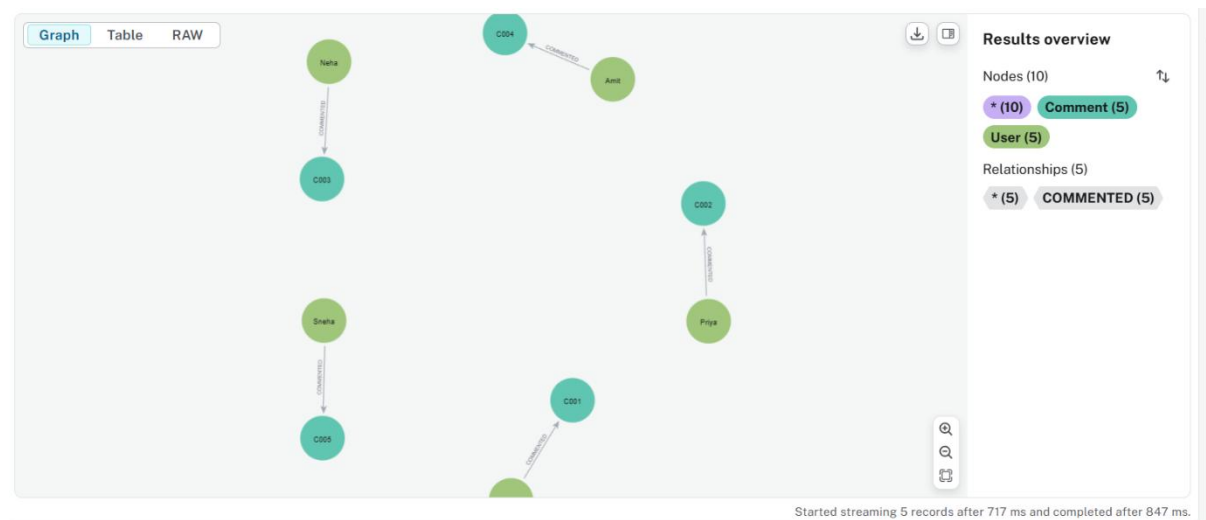
```
CREATE
(c1:Comment {id:'C001', text:'Graph databases are very useful for connected data.',
sentiment:'Positive', toxicity:0.02, likes:34, timestamp:'2026-08-20'}),
(c2:Comment {id:'C002', text:'I think Graph RAG can improve knowledge retrieval.',
sentiment:'Positive', toxicity:0.01, likes:41, timestamp:'2026-08-21'}),
(c3:Comment {id:'C003', text:'The performance depends heavily on the graph structure.',
```



```
sentiment:'Neutral', toxicity:0.03, likes:18, timestamp:'2026-08-21'}),
(c4:Comment {id:'C004', text:'Interesting perspective on community detection.',
sentiment:'Positive', toxicity:0.00, likes:29, timestamp:'2026-08-22'}),
(c5:Comment {id:'C005', text:'I disagree with this approach.', sentiment:'Negative',
toxicity:0.12, likes:9, timestamp:'2026-08-22'});
```

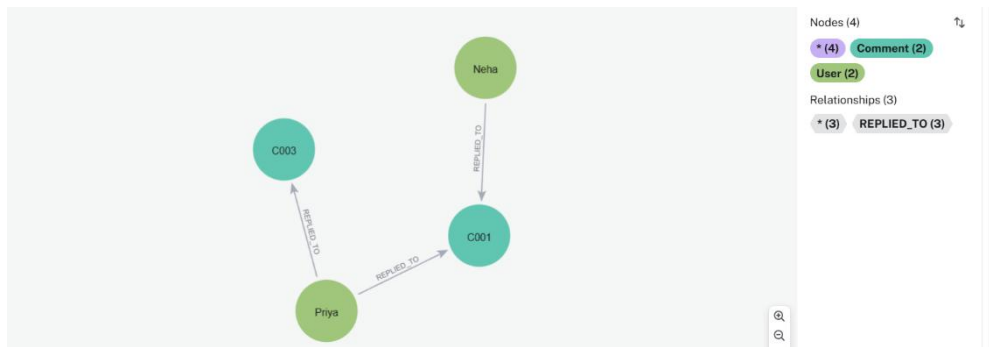
7. Connect Users to Comments

```
MATCH
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (n:User {name:'Neha'}), (a:User
{name:'Amit'}), (s:User {name:'Sneha'}),
(c1:Comment {id:'C001'}), (c2:Comment {id:'C002'}), (c3:Comment {id:'C003'}),
(c4:Comment {id:'C004'}), (c5:Comment {id:'C005'})
CREATE
(r)-[:COMMENTED {time:'10:30'}]->(c1),
(p)-[:COMMENTED {time:'11:10'}]->(c2),
(n)-[:COMMENTED {time:'12:15'}]->(c3),
(a)-[:COMMENTED {time:'13:20'}]->(c4),
(s)-[:COMMENTED {time:'14:05'}]->(c5);
```



8. Comments as Conversation Nodes

```
MATCH
(p:User {name:'Priya'}), (n:User {name:'Neha'}),
(c1:Comment {id:'C001'}), (c3:Comment {id:'C003'})
CREATE
(p)-[:REPLIED_TO]->(c1),
(n)-[:REPLIED_TO]->(c1),
(p)-[:REPLIED_TO]->(c3);
```



9. Create Topics

```

CREATE
(t1:Topic {name:'Graph Database'}),
(t2:Topic {name:'Graph RAG'}),
(t3:Topic {name:'Community Detection'}),
(t4:Topic {name:'Social Network Analysis'});

```

```

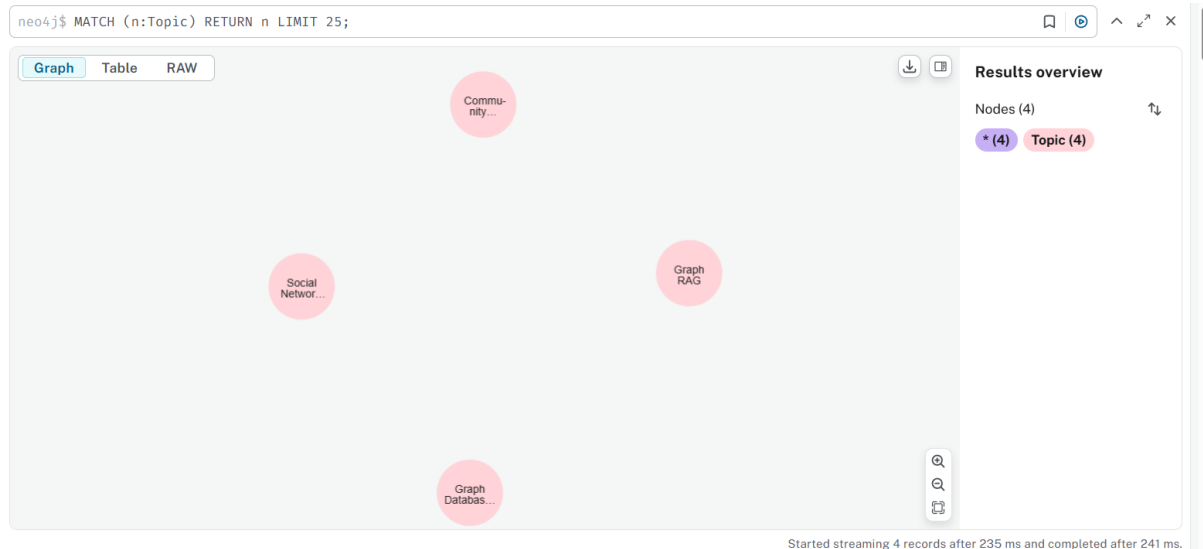
MATCH
(c1:Comment {id:'C001'}), (c2:Comment {id:'C002'}), (c3:Comment {id:'C003'}),
(c4:Comment {id:'C004'}),
(t1:Topic {name:'Graph Database'}), (t2:Topic {name:'Graph RAG'}), (t3:Topic
{name:'Community Detection'})

```

```

CREATE
(c1)-[:ABOUT]->(t1),
(c2)-[:ABOUT]->(t2),
(c3)-[:ABOUT]->(t1),
(c4)-[:ABOUT]->(t3);

```



13. Important Cypher Patterns

Operation	Pattern
Create node	CREATE (u:User {...})
Find node	MATCH (u:User {...})
Create relationship	CREATE (a)-[:FRIEND]->(b)
Find relationship	MATCH (a)-[r:FRIEND]->(b)
Any relationship	MATCH (a)-[r]-(b)
Variable-length path	MATCH path=(a)-[:FRIEND*1..3]-(b)
Return graph	RETURN a, r, b
Return path	RETURN path
Filter	WHERE condition
Count	COUNT(x)
Update	SET u.property = value
Delete relationship	DELETE r
Delete node + relationships	DETACH DELETE u

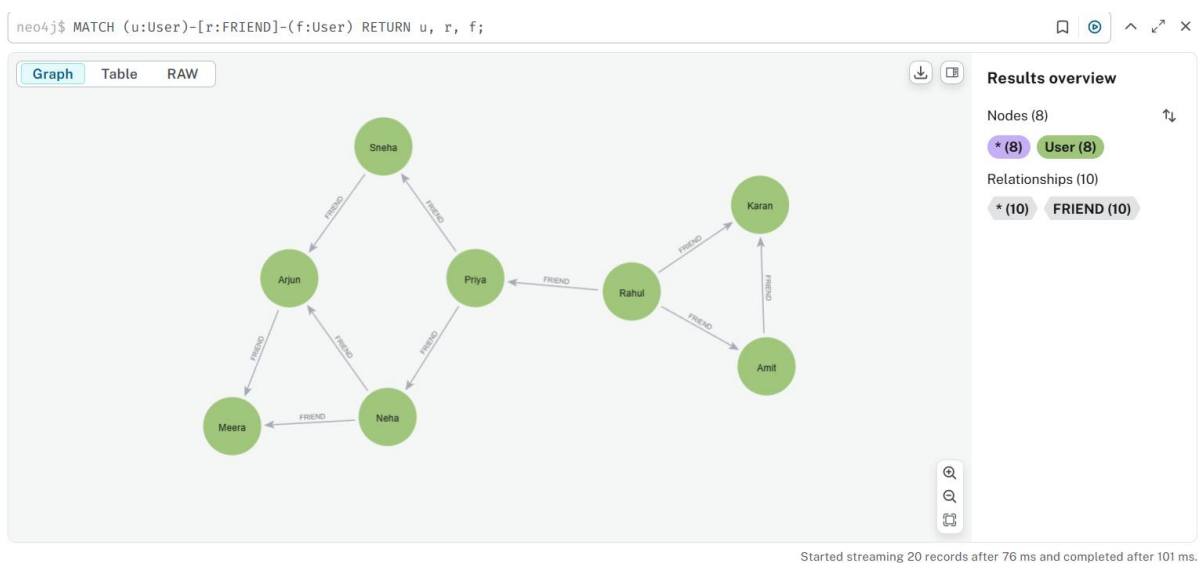
12. Graph Visualization

Purpose	Cypher	Typical result
Tabular	MATCH (u:User)-[:FRIEND]->(f) RETURN u.name, f.name	Names/rows
Graph	MATCH (u:User)-[r:FRIEND]->(f) RETURN u,r,f	Nodes + edges
Path	MATCH path=(u)-[:FRIEND*1..3]-(v) RETURN path	Connected path

Exercise 1 — Display the Complete User Network

Returns a visual network containing users and user-to-user relationships.

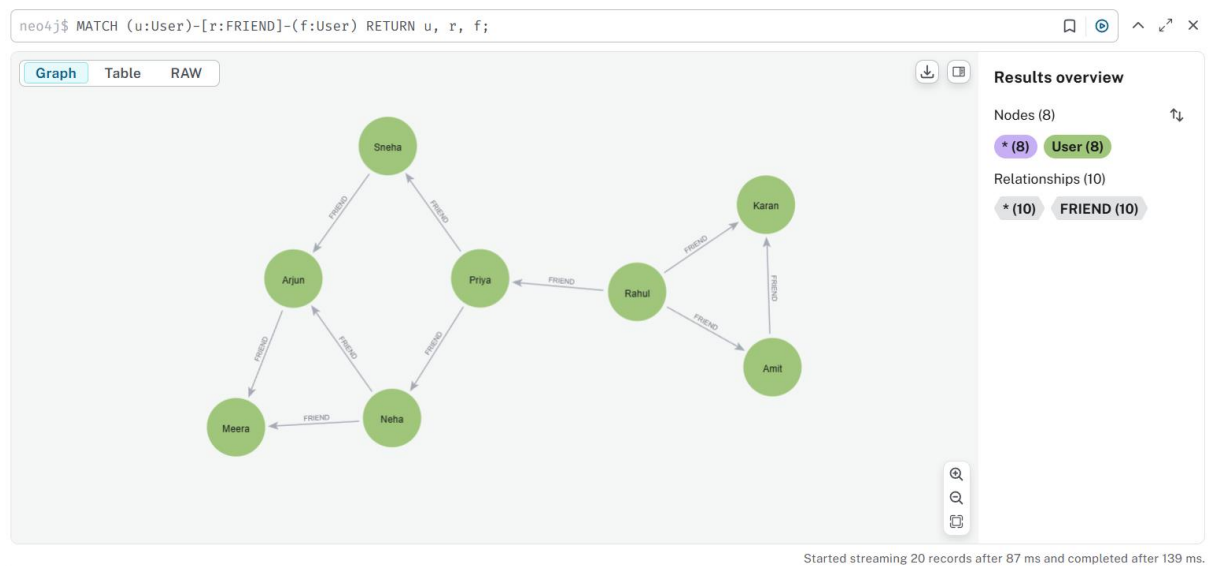
```
MATCH (u:User)-[r:FRIEND]-(f:User)
RETURN u, r, f;
```



Exercise 2 — Display Only Friendships

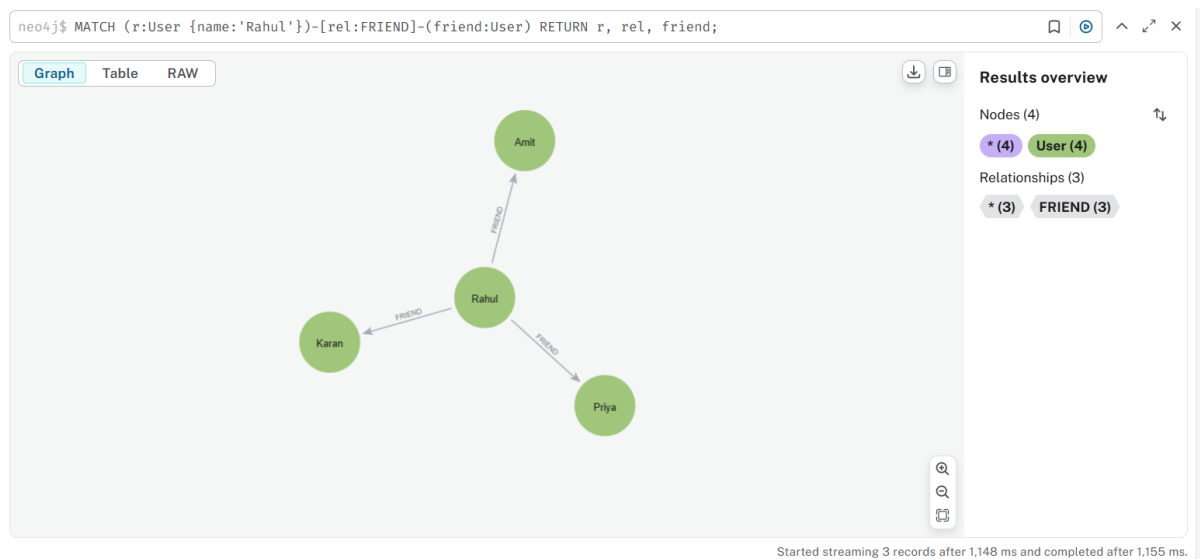
Shows the friendship graph.

```
MATCH (u:User)-[r:FRIEND]-(f:User)
RETURN u, r, f;
```



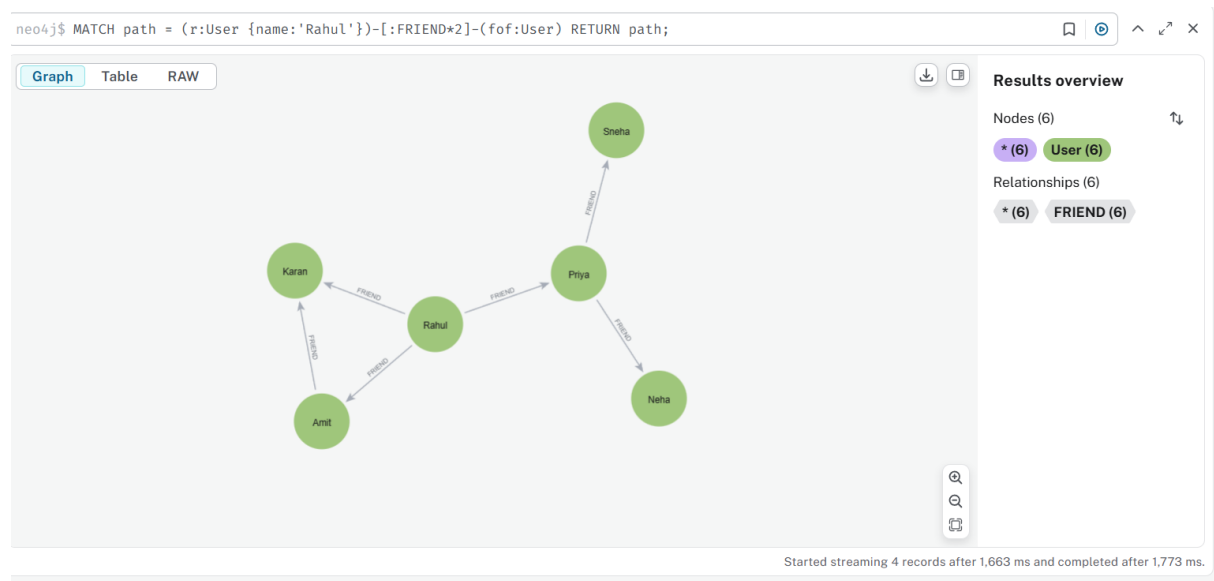
Exercise 3 — Rahul's Immediate Network

Builds Rahul's one-hop ego network.



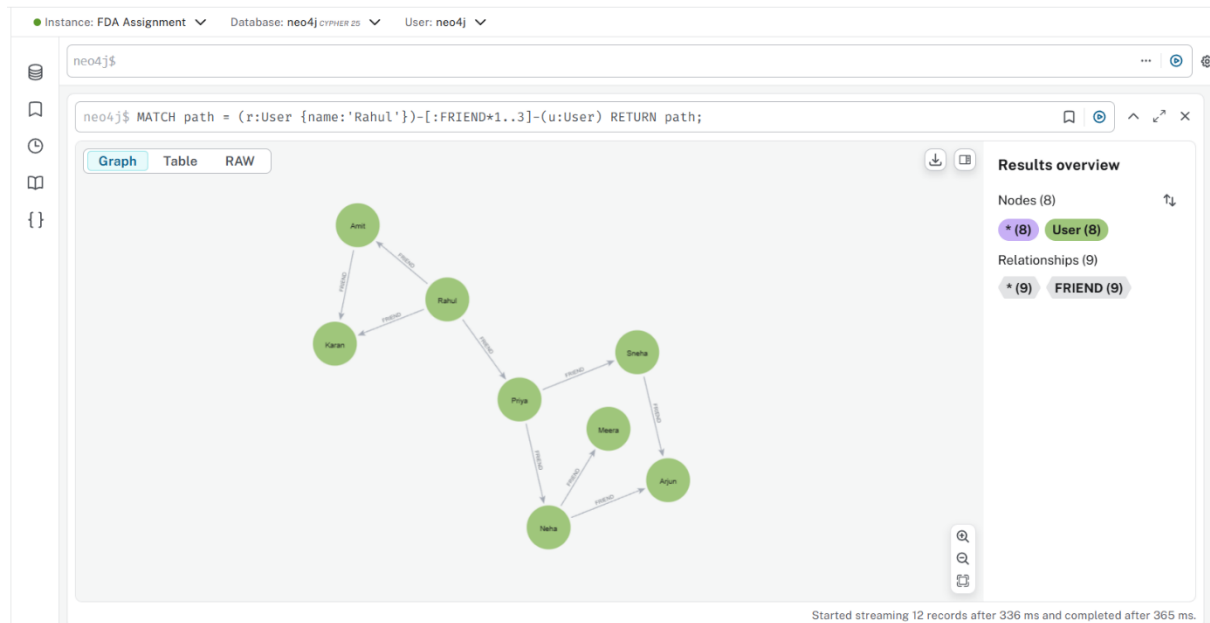
Exercise 4 — Rahul's Friends-of-Friends

Visualizes exactly two friendship hops.



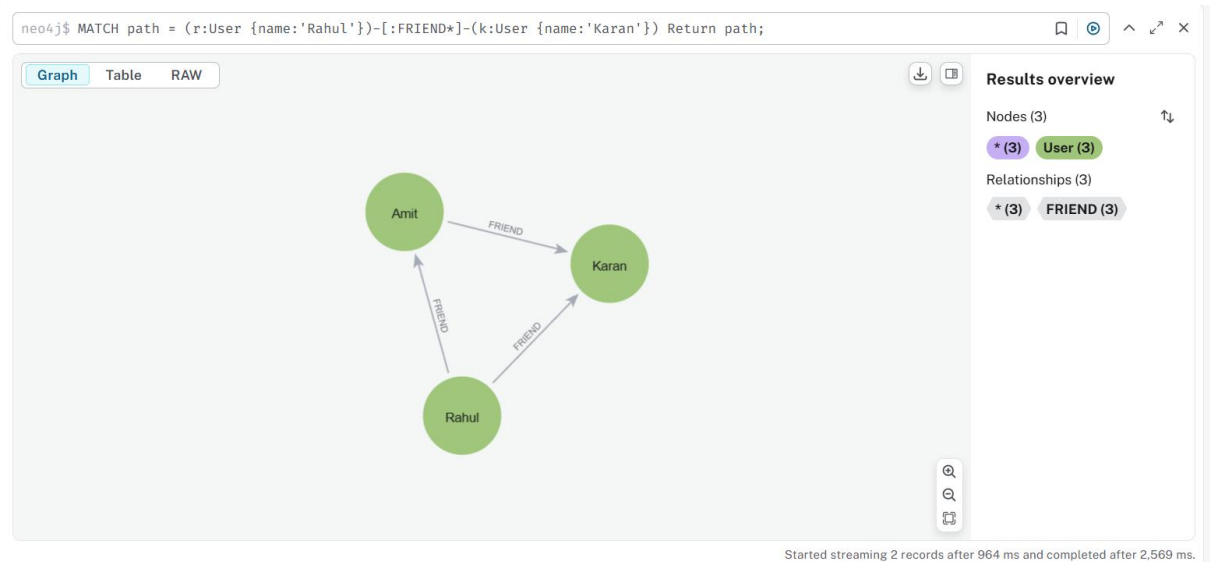
Exercise 5 — One to Three Hops

Visualizes Rahul's wider social neighborhood.



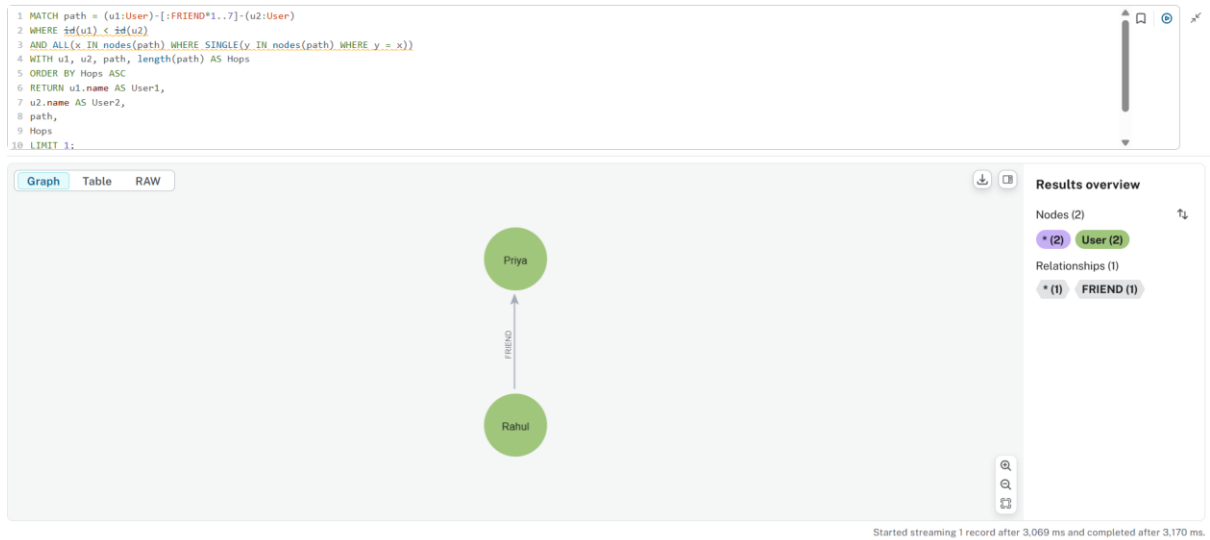
Exercise 6 — Connection Between Rahul and Karan

Returns friendship paths connecting the two users.



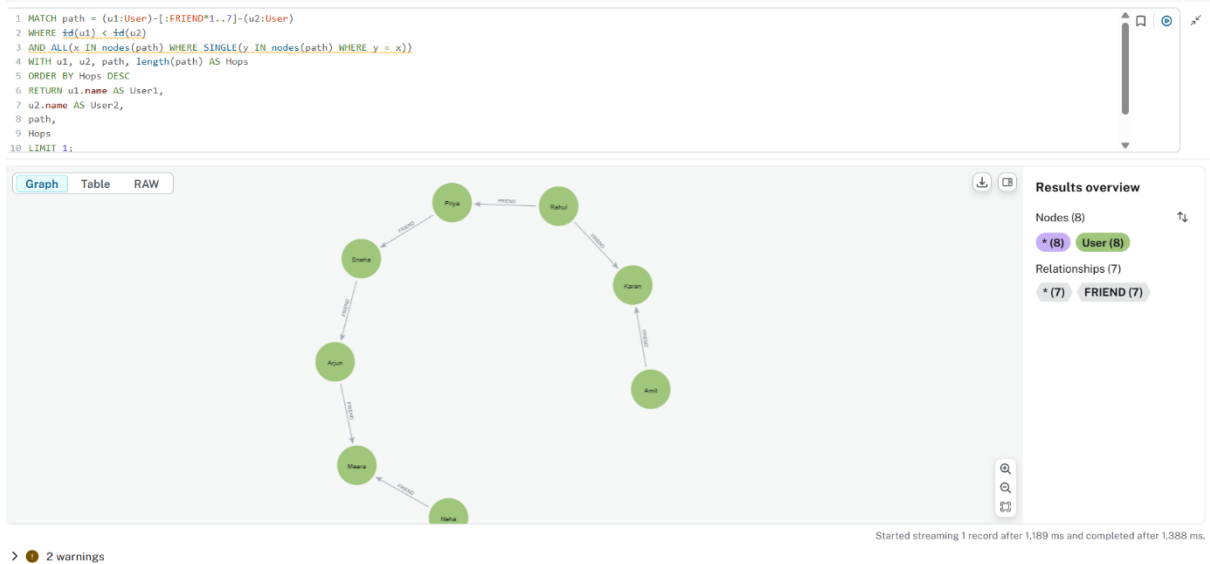
Exercise 7 — Shortest Social Connection (Any user)

Finds the shortest friendship route.



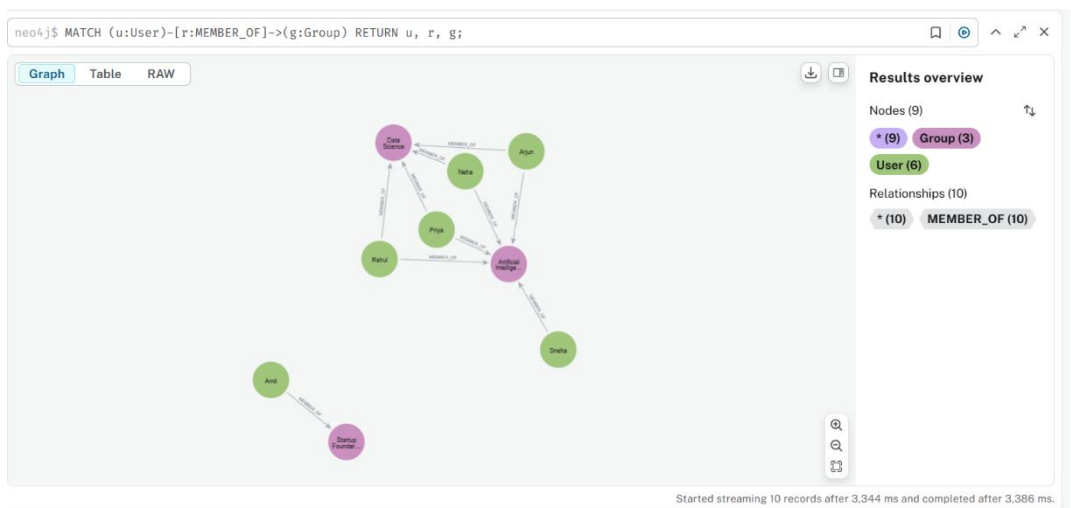
Exercise 8 — Longest Social Connection

Finds the Longest friendship route.



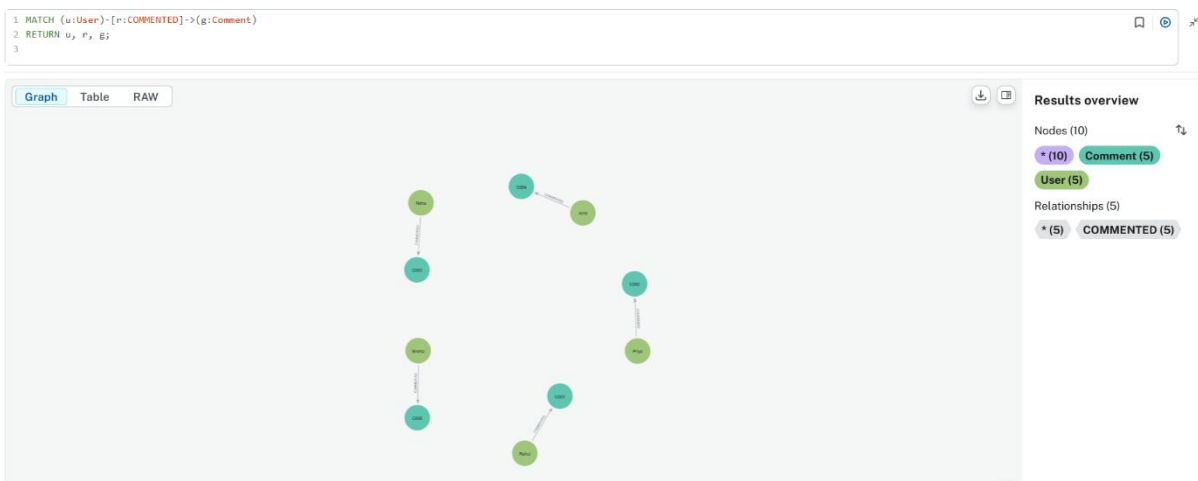
Exercise 9 — Users and Groups

Visualizes community/group membership



Exercise 10 — Users and Their Comments

Visualizes who made which comment.



```

1 MATCH (u:User)-[r:COMMENTED]->(c:Comment)
2 RETURN u.name AS User,
3 c.id AS CommentID,
4 c.text AS Comment,
5 r.time AS Time;

```

Table RAW

	User	CommentID	Comment	Time
1	"Rahul"	"C001"	"Graph database s are very usef ul for connecte d data."	"10:30"
2	"Priya"	"C002"	"I think Graph RAG can improve knowledge retri eval."	"11:10"
3	"Amit"	"C004"	"Interesting pe rspective on co mmunity detecti on."	"13:20"
4	"Neha"	"C003"	"The performanc e depends heav ily on the graph structure."	"12:15"
5	"Sneha"	"C005"	"I disagree wit h this approac	"14:05"

Exercise 11 — Comment Conversation Network

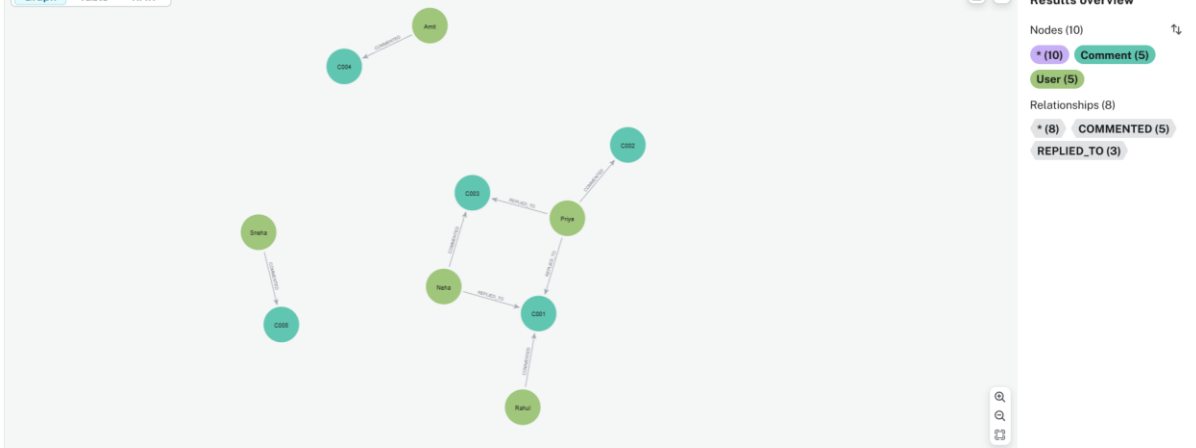
Shows users participating in a comment conversation.

```

1 MATCH (u:User)-[r:COMMENTED] REPLIED_TO -> (c:Comment)
2 RETURN u, r, c;
3

```

Graph Table RAW



Started streaming 8 records after 1,365 ms and completed after 1,407 ms.

```

1 MATCH (u:User)-[:COMMENTED|REPLIED_TO]->(c:Comment)
2 RETURN u, r, c;
3

```

u	r	c
1 (:User {profession: "Data Scientist", influenceScore: 87, city: "Pune", name: "Rahul", activityLevel: 92, age: 29})	[:COMMENTED {time: "10:30"}]	(:Comment {toxicity: 0.02, sentiment: "Positive", id: "C001", text: "Graph databases are very useful for connected data.", likes: 34, timestamp: "2026-08-20"})
2 (:User {profession: "AI Engineer", influenceScore: 82, city: "Mumbai", name: "Priya", activityLevel: 88, age: 27})	[:COMMENTED {time: "11:10"}]	(:Comment {toxicity: 0.01, sentiment: "Positive", id: "C002", text: "I think Graph RAG can improve knowledge retrieval.", likes: 41, timestamp: "2026-08-21"})
3 (:User {profession: "AI Engineer", influenceScore: 82, city: "Mumbai", name: "Priya", activityLevel: 88, age: 27})	[:REPLIED_TO]	(:Comment {toxicity: 0.02, sentiment: "Positive", id: "C001", text: "Graph databases are very useful for connected data.", likes: 34, timestamp: "2026-08-20"})
4 (:User {profession: "AI Engineer", influenceScore: 82, city: "Mumbai", name: "Priya", activityLevel: 88, age: 27})	[:REPLIED_TO]	(:Comment {toxicity: 0.03, sentiment: "Neutral", id: "C003", text: "The performance depends on the graph structure.", likes: 18, timestamp: "2026-08-21"})

Started streaming 8 records after 1.365 ms and completed after 1.365 ms

Exercise 12 — Users, Comments and Topics

Connects people to topics through comments.

```

1 MATCH (u:User)-[:COMMENTED|REPLIED_TO]->(c:Comment)-[:ABOUT]->(t:Topic)
2 RETURN u, r1, c, r2, t;
3

```

u	r1	c	r2	t
1 (:User {profession: "Data Scientist", influenceScore: 87, city: "Pune", name: "Rahul", activityLevel: 92, age: 29})	[:COMMENTED {time: "10:30"}]	(:Comment {toxicity: 0.02, sentiment: "Positive", id: "C001", text: "Graph databases are very useful for connected data.", likes: 34, timestamp: "2026-08-20"})	[:ABOUT]	(:Topic {name: "Graph Databases"})
2 (:User {profession: "AI Engineer", influenceScore: 82, city: "Mumbai", name: "Priya", activityLevel: 88, age: 27})	[:COMMENTED {time: "11:10"}]	(:Comment {toxicity: 0.01, sentiment: "Positive", id: "C002", text: "I think Graph RAG can improve knowledge retrieval.", likes: 41, timestamp: "2026-08-21"})	[:ABOUT]	(:Topic {name: "Graph RAG"})
3 (:User {profession: "Researcher", influenceScore: 91, city: "Delhi", name: "Neha", activityLevel: 95, age: 25})	[:COMMENTED {time: "12:15"}]	(:Comment {toxicity: 0.03, sentiment: "Neutral", id: "C003", text: "The performance depends heavily on the graph structure.", likes: 18, timestamp: "2026-08-21"})	[:ABOUT]	(:Topic {name: "Graph Databases"})
4 (:User {profession: "Entrepreneur", influenceScore: 79, city: "Pune", name: "Amit", activityLevel: 75, age: 35})	[:COMMENTED {time: "13:20"}]	(:Comment {toxicity: 0.0, sentiment: "Positive", id: "C004", text: "Interesting perspective on the future of AI.", likes: 12, timestamp: "2026-08-21"})	[:ABOUT]	(:Topic {name: "Community Detection"})

Started streaming 4 records after 791 ms and completed after 791 ms

Exercise 13 — Shared Topic Network

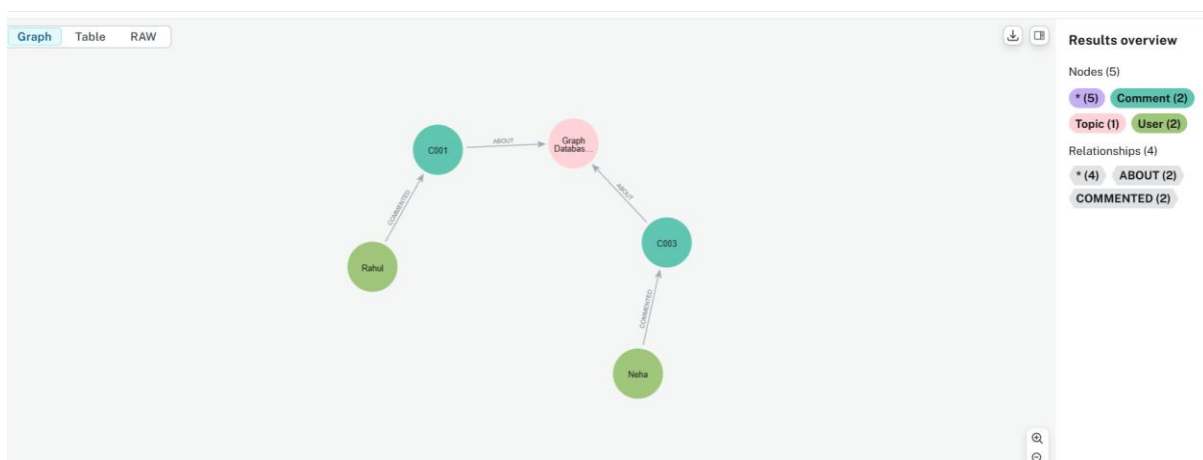
Visualizes users discussing the same topic.

```

1 MATCH (u1:User)-[r1:COMMENTED]->(c1:Comment)-[r2:ABOUT]->(t:Topic)
2 MATCH (u2:User)-[r3:COMMENTED]->(c2:Comment)-[r4:ABOUT]->(t)
3 WHERE u1 <> u2
4 RETURN u1, r1, c1, r2, t, r4, c2, r3, u2;

```

	c1	r2	t	r4	c2
: "12:15"}}	(:Comment {toxicity: 0.03, sentiment: "Neutral", id: "C003", text: "The performance depends heavily on the graph structure.", likes: 18, timestamp: "2026-08-21"})	[[:ABOUT]]	(:Topic {name: "Graph Databases", id: "T001"})	[[:ABOUT]]	(:Comment {toxicity: 0.02, sentiment: "Positive", id: "C001", text: "Graph databases are very useful for connected data.", likes: 34, timestamp: "2026-08-20"})
: "10:30"}}	(:Comment {toxicity: 0.02, sentiment: "Positive", id: "C001", text: "Graph databases are very useful for connected data.", likes: 34, timestamp: "2026-08-20"})	[[:ABOUT]]	(:Topic {name: "Graph Databases", id: "T001"})	[[:ABOUT]]	(:Comment {toxicity: 0.03, sentiment: "Neutral", id: "C003", text: "The performance depends heavily on the graph structure.", likes: 18, timestamp: "2026-08-21"})



Exercise 14 — High-Toxicity Interaction

Visualizes conversations around high-toxicity comments.

```

neo4j$ MATCH (u:User)-[r:COMMENTED|REPLIED_TO]->(c:Comment) WHERE c.toxicity > 0.10 RETURN u, r, c;

```

u	r	c
(:User {profession: "Product Manager", influenceScore: 86, city: "Mumbai", name: "Sneha", activityLevel: 84, age: 28})	[[:COMMENTED] {time: "14:05"}]	(:Comment {toxicity: 0.12, sentiment: "Negative", id: "C005", text: "I disagree with this approach.", likes: 9, timestamp: "2026-08-22"})

Started streaming 1 record after 868 ms and completed after 874 ms.

Exercise 15 — Friend Recommendation Graph

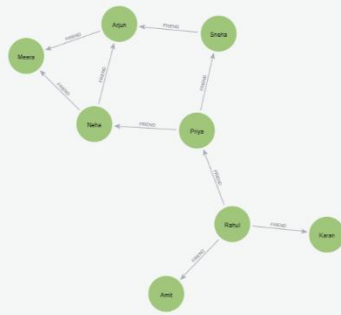
Returns the graph structure behind a friend recommendation.

```

1 MATCH (u:User)-[r1:FRIEND]-(mutual:User)-[r2:FRIEND]-(candidate:User)
2 WHERE u <> candidate
3 AND NOT (u)-[:FRIEND]-(candidate)
4 RETURN u, r1, mutual, r2, candidate;

```

Graph Table RAW



Download

Results overview

Nodes (8)

* (8) User (8)

Relationships (9)

* (9) FRIEND (9)

Exercise 16 — Complete Social Interaction Graph

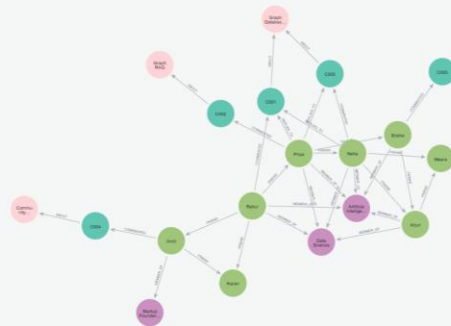
Visualizes all currently modeled relationships

```

1 MATCH (n)-[r]-(m)
2 RETURN n, r, m;

```

Graph Table RAW



Download

Results overview

Nodes (19)

* (19) Comment (5)

Group (3) Topic (3)

User (8)

Relationships (32)

* (32) ABOUT (4)

COMMENTED (5)

FRIEND (10)

MEMBER_OF (10)

REPLIED_TO (3)